

Total No. of Questions :5]

SEAT No. :

PE4197

[Total No. of Pages : 2

[6554]-2001

F.Y. M.C.A. (Management)

JPR551MJ-IT 21 : JAVA PROGRAMMING

(2024 Pattern) (Semester-II)

Time :2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Figures to the right indicate full marks.*

Q1) Attempt any two

- a) Define class student with suitable attributes and methods. Display the student details. **[5]**
- b) Create an abstract class shape with method area(). Inherit this class for creating square class. Display area of square. **[5]**
- c) Write a function using lambda expression to calculate to cube of a given number. **[5]**
- d) Explain use of final keyword for class, data members, methods. **[5]**

Q2) Attempt any two

- a) Explain Thread Life Cycle with suitable diagram. **[5]**
- b) Accept age from user. If age is negative throw Invalid Age exception. (user defined) **[5]**
- c) Demonstrate synchronized keyword with example. **[5]**
- d) Write a program to demonstrate Hashset. Create a Hashset and perform operations add and remove elements. **[5]**

Q3) Attempt any one

- a) Create a Servlet which checks username and password, if it is correct, display message "Successful login," else display appropriate error message (without JDBC) **[10]**
- b) Demonstrate session management using cookies. **[10]**

P.T.O.

Q4) Attempt any one

- a) Create a jsp page to accept roll no. of a student, update mobile number in the student table. Display the updated details. Assume suitable structure. [10]
- b) Explain any 5 implicit objects of JSP. [10]

Q5) Attempt any one

- a) Create a spring mvc form to read registration details for placement drive with spring validation and display it. [10]
- b) Explain Spring mvc flow and Inversion of control (IoC) [10]